

Deep Learning-based Motion Models for Multi-Target Tracking of Inland Vessels using AIS Data

Motivation

Autonomous surface vessels must perceive and understand their environment to operate safely. A key part of this is predicting how surrounding vessels will move. Multi-target tracking (MOT) systems perform this task by associating detections over time and maintaining object trajectories. While onboard sensors such as radar, LiDAR, and cameras provide local observations, the Automatic Identification System (AIS) offers large-scale motion data containing vessel position, velocity, and identity. This rich source of real trajectories enables data-driven learning of vessel motion behavior. In contrast, classical models like constant velocity or acceleration assume simplified kinematics and cannot represent complex maneuvers or environmental influences. Using AIS data, deep learning-based motion models could learn more realistic vessel dynamics and improve prediction performance in maritime tracking.

Problem definition

Usually, multi-target tracking uses the following scheme: Predict the state into the future using a motion model, associate the correct measurement (detection) to it, correct the predicted state based on the measurement (sensor model). In current maritime applications, the motion model is typically a simple constant acceleration model. This provides only a coarse representation of vessel kinematics and assumes uncorrelated Gaussian noise for deviations. Such models cannot capture nonlinear maneuvers, course changes, or varying hydrodynamic responses. With the availability of large AIS datasets ([Vessel-Tracker](#)), it becomes feasible to learn vessel motion directly from data. Since AIS messages contain timestamped positions, velocities, and vessel identifiers, sequences of vessel trajectories can be extracted and used to train deep learning-based motion models such as Long Short-Term Memory (LSTM) networks, extended architectures like xLSTM, or Transformer-based sequence models. These models can learn temporal dependencies and motion patterns over time, effectively encoding typical vessel dynamics in different operational scenarios. The proposed work aims to train and evaluate such learning-based motion models on AIS-derived trajectory data and to compare their predictive performance to classical analytical models such as constant velocity, constant acceleration, and constant turn rate models. The goal is to quantify how much learned motion dynamics improve trajectory prediction accuracy and to assess their potential integration into future maritime multi-target tracking frameworks.

Tasks

- Literature Review on deep learning-based methods for trajectory prediction (LSTM, Transformer, etc.)
- Identify and implement adequate methods and compare to classical motion models
- Test the method and compare on real world and simulated datasets

Requirements

- Programming skills in Python (PyTorch or TensorFlow) or C++
- Experience with VS Code, Git, Linux is beneficial
- Knowledge about deep learning with sequential data and Kalman Filter is beneficial

Benefits

- Independent practical work with close scientific supervision at an AI-Institute
- Real-world (& simulated) dataset
- We aim to publish the results at a renowned conference

Supervisor

Tim Rehbronn, M. Sc.

Gebäude: Cluster Produktionstechnik 3b 01.036.

+49 241 80-22296

t.rehbronn@irt.rwth-aachen.de

Institute of Automatic Control

RWTH Aachen University

Campus-Boulevard 30

52074 Aachen

www.irt.rwth-aachen.de

